RESOLVE: Rare Event Surrogate Likelihood for Gravitational Wave Paleontology Parameter Estimation

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Abstract

The first detection of gravitational waves, recognized by the 2017 Nobel Prize in Physics, has opened up a new research field: gravitational-wave paleontology. When massive stars evolve into black holes and collide, they create gravitational waves that propagate through space and time. These gravitational-waves, now detectable on Earth, act as fossils tracing the histories of the massive stars that created them. Estimating physics parameters of these massive stars from detected gravitational-waves is a parameter estimation task, with the primary difficulty being the extreme rarity of collisions in simulated binary black holes. This rarity forces researchers to choose between prohibitively expensive simulations or accepting substantial statistical variance. In this work, we present RESOLVE, a rare event surrogate model that leverages polynomial chaos expansion (PCE) and Bayesian MCMC to emulate this rare formation efficiency. Our experimental results demonstrate that RESOLVE is the only surrogate model that achieves proper statistical coverage, while effectively learning the underlying distribution of each physics parameter. We construct a likelihood function incorporating both the emulated formation efficiency and LIGO's gravitational wave observations, which we then minimize to produce community-standard credible intervals for each physics parameter. These results enable astronomers to gain deeper insights into how the universe transformed from simple gases into the complex chemical environment that eventually made life possible.

1 Introduction

On September 14, 2015, Laser Interferometer Gravitational-Wave Observatory (LIGO) announced the detection of GW150914, marking humanity's first ever direct observation of gravitational waves [Abbott et al., 2016]. This groundbreaking result was swiftly endorsed by the 2017 Nobel Prize in Physics. Gravitational waves were first theorized by Albert Einstein in 1916 [Einstein, 1916, 1918]

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as part of his general relativity framework, which predicted that massive objects distort the space and time around them and when they accelerate, they create "ripples in spacetime": gravitational waves that propagate through the universe at the speed of light.

The astronomical events that generate detectable gravitational waves involve the so-called **binary systems**: when two massive stars evolve into two black holes (or neutron stars), they will orbit around each other at accelerated speed and eventually collide. These collisions release tremendous energy, producing gravitational-waves that can be detected with observatories such as LIGO Abbott et al. [2016], VIRGO Acernese et al. [2014], and KAGRA Akutsu et al. [2019] [1]. Because it can take millions to billions of years for binary black holes (BBH) to collide and be detected through gravitational waves, these signals can act as "astronomical fossils", tracing the histories of millions of their progenitor stars across cosmic time. This new frontier is called Gravitational Wave Paleontology.

Learning the properties of massive stars in binary system from detected gravitational waves can be described as a parameter estimation task. Suppose the massive stars exhibit certain properties that can be characterized by physics parameters θ: the first step involves a theoretical forward model that accepts θ and outputs whether a pair of black holes will be formed to create a gravitational wave. This output can then be compared to the actual gravitational wave observations y_{obs} to draw a confidence limit or credible intervals over θ. Astronomers can then use these intervals to test existing theories of the evolution of stars and the underlying physical processes such as stellar outflows, enrichment, nuclear fusion and supernovae, leading to an improved understanding of how the universe transformed from simple gases into the complex chemical environment that eventually made life possible.

The key challenge lies in the theoretical forward model, which, in the astronomy community, is encoded within a well-established software framework Compact Object Mergers: Population Astrophysics and Statistics (COMPAS) Riley, J. et al. [2022] Stevenson et al. [2017] Vigna-Gómez et al. [2018]. Currently, COMPAS simulations are limited by uncertainty and high computation cost, making it impossible to explore the full parameter space of θ. Suppose N binary systems are simulated in COMPAS under physics parameter $\boldsymbol{\theta}$, but only m of them form black hole pairs that eventually collide and produce observable gravitational wave signals. The **formation efficiency** is thereby calculated as $\epsilon = m/N$. Given that ϵ is intrinsically small, millions of binary systems must be simulated to collect only a few collisions, making each evaluation of ϵ computationally prohibitive. Consequently, learning a mapping between $\boldsymbol{\theta}$ and ϵ becomes intractable under traditional approaches.

In this work, we developed the RESOLVE model, which includes a novel rare event surrogate model that, after training, maps $\boldsymbol{\theta}$ to $\hat{\epsilon}$ without actually running the time-consuming COMPAS simulation. We can further convert $\hat{\epsilon}$ into an expected gravitational wave rate $\hat{y}$ to be compared to the observed rate y_{obs}. The last step involves constructing a likelihood function $L(y_{obs}|\hat{y})$ and minimizing it with Bayesian inference algorithms to obtain credible interval over $\boldsymbol{\theta}$.

2 Related Works

It is commonplace across disciplines to use machine learning techniques as surrogates for computationally-expensive simulations [Donnelly et al., 2024, Würth et al., 2023, Oldenburg et al., 2022, Shibata et al., 2022, Sasanapuri et al., 2025, Z. Fu et al., 2024]. Most closely related to this work is the RESuM model by Schuetz et al. [2025], which addresses similar rare event design problems in physics detector design optimization using a Multi-Fidelity Gaussian Process (MFGP) surrogate model. The key insight is the adoption of Conditional Neural Processes (CNP, Garnelo et al. [2018]) to smooth out the discreteness of the rare design metric, providing additional prior information to the MFGP surrogate model. In this work, CNP was also adopted for similar purpose, but we designed a novel surrogate model based on Polynomial Chaos Expansion (PCE) Xiu and Karniadakis [2002] for BBH simulation.

Another related field is rare event simulation and modeling in reliability engineering. The rare event problem here focuses on emulating the extremely low failure probabilities P_f. Since direct Monte Carlo simulation becomes intractable as P_f approaches zero, specialized techniques including adaptive sampling [Bucher, 1988], surrogate-based methods [Li and Xiu, 2010, Li et al., 2011], sequential importance sampling [Papaioannou et al., 2016], and multi-fidelity approaches [Peherstorfer et al., 2016, 2018], multilevel sampling [Wagner et al., 2020] and ensemble Kalman filters [Wagner et al.,

[1] See Appendix E for more details

2022] are developed. While Adaptive Importance Sampling (AIS) method has the potential to solve the rare event problem, the benchmarking study in Schuetz et al. [2025] showed that it is difficult to implement yet still underperform the CNP approach. It is worth noting that an AIS method called STROOPWAFFLE Broekgaarden et al. [2019] has been inplemented in COMPAS for simulations and can improve the initial condition sampling but has not been used as a surrogate model.

Lastly, in particle physics and astrophysics, parameter estimation relies extensively on both frequentist and Bayesian statistical inference. The chi-square fitting method, a cornerstone of frequentist inference, has been employed for decades to extract physical parameters from experimental data, particularly in collider experiments Cowan et al. [2011] and neutrino oscillation studies Feldman and Cousins [1998]. Complementarily, Bayesian inference utilizing Markov Chain Monte Carlo (MCMC) methods has gained significant attention due to its ability to incorporate prior knowledge Trotta [2008], Feroz et al. [2009], Straub et al. [2024]. These methods have proven especially valuable in cosmological parameter estimation Collaboration et al. [2020], dark matter searches Aprile et al. [2023], Aalbers et al. [2024], and the search for rare decay processes Abe et al. [2023], Agostini et al. [2020], Arnquist et al. [2023]. In RESOLVE, we inherited the conventional approaches from physics and astronomy to construct the likelihood function and draw credible interval. We also provided a more detailed description about Bayesian inference in Physics in Appendix F and G

3 The RESOLVE Model

RESOLVE is a rare event surrogate likelihood constructed to solve the rare event problem Gravitational Wave Paleontology parameter estimation. This section aims to adapt the Rare Event Design problem in [Schuetz et al., 2025] to BBH collision simulation.This section provides a comprehensive overview of the design and training of RESOLVE model. The first step involves training a Conditional Neural Process (CNP) model to produce CNP scores β_{ji} for all simulated events. Section 3.2 describes our newly developed surrogate model based on the Multi-Fidelity Bayesian Polynomial Chaos Expansion (MF-BPCE). This model is trained using both averaged CNP scores and raw formation efficiency measurements. After training, the surrogate model maps $\boldsymbol{\theta}$ to estimated formation efficiency with quantified uncertainty: $\boldsymbol{\theta} \rightarrow \hat{\epsilon} \pm \hat{\sigma}$. Finally, Section 3.3 describes the construction of a likelihood function that connects the emulated efficiency values $\hat{\epsilon}$ with observed gravitational wave signals y_{obs}.

3.1 The Rare Event Problem

Consider a scenario where we run J simulation trials indexed by j; each simulation trial contains N binary systems indexed by i. For each system i in trial j, the outcome X_{ji} is either 1, meaning that the ith pair of stars formed black hole pairs that collided, merge and emitted a detectable gravitational wave, or 0, meaning that the binary system did not form BBH mergers. If m black hole merger formed in N simulated binary systems, the formation efficiency ϵ for trial j can be defined as:

$$\epsilon_j = \frac{m}{N} = \frac{\sum_{i=1}^{N} X_{ji}}{N} \tag{1}$$

Let $\boldsymbol{\theta}$ denote the physics parameters of interest and $\boldsymbol{\phi}$ the nuisance parameters. The outcome of ϵ may depend on both $\boldsymbol{\theta}$ and $\boldsymbol{\phi}$. All binary systems in the same trial j share the same values of the parameters of interest $\boldsymbol{\theta}_j$, but have different nuisance parameters $\boldsymbol{\phi}_{ji}$. Each trial can be either Low-Fidelity (LF) or High-Fidelity (HF), depending on the number of binary systems simulated ($N_{HF} \gg N_{LF}$). The LF simulation suffers from high statistical variance since N_{LF} is small, but the lower computational cost enables broader exploration of the $\boldsymbol{\theta}$ parameter space. Meanwhile, the more expensive HF simulations allow us to obtain a ϵ that is a better estimate of the true formation efficiency. In the following context, we define ϵ_{Raw} as the formation efficiency calculated directly from m/N.

The number of formations m follows a binomial distribution with the probability $e(\boldsymbol{\theta}, \boldsymbol{\phi_i})$. The high computational cost arises from the Rare Event Condition: since $e(\boldsymbol{\theta}, \boldsymbol{\phi_i})$ is intrinsically very small, the number of forming black holes mergers m becomes negligible compared to N ($m \ll N$). This means m can be approximated by a Poisson distribution as $m \sim \text{Poisson}\left(N\bar{e}(\boldsymbol{\theta})\right)$. Where $\bar{e}(\boldsymbol{\theta})$ is the expected merger formation probability marginalized over the nuisance parameter $\boldsymbol{\phi}$:

$$\bar{e}(\boldsymbol{\theta}) = \int e(\boldsymbol{\theta}, \boldsymbol{\phi}) g(\boldsymbol{\phi}) d\boldsymbol{\phi} = \prod_i \int e(\boldsymbol{\theta}, \boldsymbol{\phi}_i) g(\boldsymbol{\phi}_i) d\boldsymbol{\phi}_i \tag{2}$$

$g(\phi)$ is a predefined distribution where we sampled ϕ_i from. However, the analytical form of $e(\theta, \phi)$ is unavailable, and direct evaluation of $e(\boldsymbol{\theta}, \boldsymbol{\phi})$ is impossible either. We only have access to X_i, generated via:

$$X_i = \text{Bernoulli}[e(\boldsymbol{\theta}, \boldsymbol{\phi})]. \tag{3}$$

When N becomes large, according to the central limit theorem, the formation efficiency ϵ will follow a normal distribution with symmetric and well-defined statistical uncertainties $\bar{e}(\theta)/N$; As $N \to +\infty$, ϵ will asymptotically approximate $\bar{e}(\theta)$ with statistical uncertainties approaching 0. But when N is small, m has a higher variance; $\epsilon = \frac{m}{N}$ can no longer be approximated with a normal distribution. In other words, ϵ will only take on a discrete set of values, $\epsilon \in \left\{ \frac{0}{N}, \frac{1}{N}, \dots, \right\}$.

To mitigate the high variance and discreteness of ϵ, Schuetz et al. [2025] leveraged a Conditional Neural Process (CNP) algorithm to transform the binary outcomes of X_{ji} (formation or no formation) into continuous floating-point scores β_{ji} between 0 and 1. In this framework, the CNP learns a Gaussian approximation to the posterior of the latent function:

$$q(e(\boldsymbol{\theta}, \boldsymbol{\phi})) \approx \mathcal{N}(\mu_{\text{NN}}(\boldsymbol{\theta}, \boldsymbol{\phi}), \sigma^2_{\text{NN}}(\boldsymbol{\theta}, \boldsymbol{\phi})),$$

trained by minimizing the expected negative log-likelihood under the Bernoulli model:

$$\mathcal{L} = -\sum_{j=1}^{J} \sum_{i=1}^{N} \log \int \text{Bernoulli}(X_{ji} \mid e(\boldsymbol{\theta}_j, \boldsymbol{\phi}_{ji}))\, q(e(\boldsymbol{\theta}, \boldsymbol{\phi}))\, de.$$

The predicted mean μ_{NN} serves as a smoothed estimate $\beta_{ji} \approx e(\boldsymbol{\theta}_j, \boldsymbol{\phi}_{ji})$. These scores are averaged across instances to produce a smooth surrogate $\epsilon^{(f_0)}_{\text{CNP}}$

$$\epsilon_{\text{CNP}} = \bar{e}(\boldsymbol{\theta}_j) = \frac{1}{N} \sum_{i=1}^{N} \beta_{ji}. \tag{4}$$

which replaces the direct empirical estimate m/N, providing a denoised input for the downstream surrogate modeling.

3.2 Multi-Fidelity Bayesian Polynomial Chaos Expansion

The model integrates sparse Polynomial Chaos Expansion (PCE) with hierarchical Bayesian inference to efficiently propagate uncertainty across both HF and LF simulation trials. The goal is to emulate the raw formation efficiency $\epsilon^{HF}_{Raw} = m/N$ evaluated on HF simulations, using both LF and optionally medium-fidelity (MF) data, thus reducing computational cost while maintaining accuracy and uncertainty quantification capabilities. Figure 1 shows a schematic overview of the model approach.

Let $\boldsymbol{\theta} \in \mathbb{R}^l$ be the input parameter vector and let $\epsilon^{(f_i)}(\boldsymbol{\theta}) \in \mathbb{R}$ denote the model output at the fidelity level $f_i \in \mathcal{F} = \{f_0, f_1, \dots, f_n\}$, where f_0 is the lowest and f_n the highest fidelity. The goal is to construct a surrogate for $\hat{\epsilon}^{(f_n)}(\boldsymbol{\theta})$ that uses all available fidelity data and quantifies the uncertainty in its predictions. $\hat{\epsilon}^{(f_0)}(\boldsymbol{\theta})$ can be expanded into a finite series of multivariate orthogonal polynomials:

$$\hat{\epsilon}^{(f_0)}(\tilde{\boldsymbol{\theta}}) = \sum_{k=1}^{d^{(f_0)}} c_k^{(f_0)} \Psi_k(\tilde{\boldsymbol{\theta}}),$$

where $\tilde{\boldsymbol{\theta}} \in [-1, 1]^l$ is the normalized input, and $\Psi_k(\cdot)$ are multivariate Legendre basis functions formed as tensor products of 1D polynomials:

$$\Psi_k(\tilde{\boldsymbol{\theta}}) = \prod_{i=1}^{d^{(f_0)}} L_{k_i}(\tilde{\boldsymbol{\theta}}_i),$$

with multi-index $k \in \mathbb{N}_0^{d^{(f_0)}}$ satisfying $\|k\|_1 \leq d^{(f_0)}$, and $d^{(f_0)}$ being the maximum total degree.

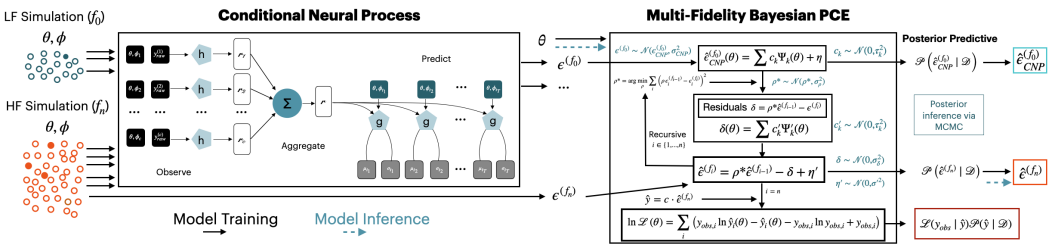


Figure 1: Overview of the RESOLVE framework. The left side illustrates the CNP used for modeling both LF and HF simulation data. The CNP aggregates nuisance parameters $\boldsymbol{\phi}_i$ and parameters of interest $\boldsymbol{\theta}$ from LF and HF simulations to produce $\epsilon_{\text{CNP}}^{(f_0)}$, which, together with HF formation efficiency $\epsilon_{Raw}^{(f_n)}$, serve as inputs to the surrogate model. The right side shows the multi-fidelity PCE that combines LF predictions $\epsilon_{\text{CNP}}^{(f_0)}$ and HF simulation estimates $\epsilon_{Raw}^{(f_n)}$ providing a surrogate $\hat{\epsilon}^{(f_n)}$ for the HF formation efficiency .

For the multi-fidelity surrogate construction, we adopt an autoregressive model inspired by the work Kennedy and O'Hagan [2000]. In this framework, each fidelity level $f_i \in \{f_0, \ldots, f_n\}$ is assumed to have a linear (scalar) relationship with the prediction from the preceding fidelity level f_{i-1}, plus an additive discrepancy term modeled via a PCE as well. Formally, the model is defined recursively as

$$\hat{\epsilon}^{(f_i)}(\tilde{\boldsymbol{\theta}}) = \rho^{(f_i)} \hat{\epsilon}^{(f_{i-1})}(\tilde{\boldsymbol{\theta}}) + \sum_{k=1}^{d^{(f_i)}} c_k^{(f_i)} \Psi_k(\tilde{\boldsymbol{\theta}}), \quad \text{for } f_i = f_0, \ldots, f_n. \tag{5}$$

where higher-fidelity predictions $\hat{\epsilon}^{(f_i)}(\tilde{\boldsymbol{\theta}})$ are expressed as corrections to the scaled lower-fidelity predictions with $\rho^{(f_i)} \in \mathbb{R}$ being a scalar scaling parameter capturing the linear relationship between fidelities f_i and f_{i-1}. The discrepancy term $\sum_k c_k^{(f_i)} \Psi_k(\tilde{\boldsymbol{\theta}})$ captures residual structure not accounted for by the scaled prediction $\rho^{(f_i)} \hat{\epsilon}^{(f_{i-1})}(\tilde{\boldsymbol{\theta}})$, allowing the model to learn fidelity-specific corrections.

This autoregressive multi-fidelity structure naturally enables coherent uncertainty propagation across fidelity levels by explicitly modeling the relationships between low- and high-fidelity outputs within a probabilistic framework. To formalize this, we adopt a Bayesian linear regression formulation that allows us to quantify and propagate epistemic uncertainty in the model coefficients and discrepancy terms. Specifically, we treat the model output $\hat{\epsilon}^{(f_i)}$ at fidelity level f_i as a noisy observation of a PCE

$$\hat{\epsilon}^{(f_i)} \sim \mathcal{P}\left(\hat{\epsilon}^{(f_i)}(\tilde{\boldsymbol{\theta}}), \boldsymbol{\eta}^{(f_i)}\right),$$

where $\hat{\epsilon}^{(f_i)}(\tilde{\boldsymbol{\theta}})$ denotes the deterministic PCE approximation at input $\boldsymbol{\theta}$, and $\boldsymbol{\eta}^{(f_i)}$ are parameters governing the noise or discrepancy distribution at fidelity level f_i. The likelihood function $\mathcal{P}$ can be chosen flexibly, depending on the characteristics of the data.

Benchmarking results in Schuetz et al. [2025] demonstrated that including the CNP scores enhances the performance of Multi-Fidelity Gaussian Process (MFGP) surrogate models. In this work, we adopted the same CNP structure as Schuetz et al. [2025] for the gravitational wave analysis. It acts as a learned prior for the lowest-fidelity level, and its predictions are passed into the autoregressive multi-fidelity model:

$$\hat{\epsilon}^{(f_0)} \sim \mathcal{N}\left(\epsilon_{\text{CNP}}^{(f_0)}, \sigma_{\text{CNP}}^2\right).$$

The coefficients of the PCE, denoted $\{c_k^{(f_i)}\}$, define the contribution of each basis function at fidelity level f_i, while a scaling parameter $\rho^{(f_i)}$ modulates the influence of the lower-fidelity prediction. These parameters are treated as latent parameters with prior distributions:

$$\{c_k^{(f_i)}\} \sim \mathcal{P}_{\text{c}}, \quad \boldsymbol{\eta}^{(f_i)} \sim \mathcal{P}_{\text{n}}, \quad \rho^{(f_i)} \sim \mathcal{P}_{\text{s}}.$$

The prior on $\{c_k^{(f_i)}\}$ regularizes the expansion to avoid overfitting. The specific choice of priors $\mathcal{P}_\mathrm{c}, \mathcal{P}_\mathrm{n}, \mathcal{P}_\mathrm{s}$ can be adapted to reflect prior knowledge, modeling needs, or regularization preferences.

Given data $\mathcal{D}^{(f_i)} = \{(\boldsymbol{\theta}_j, \epsilon_j^{(f_i)})\}_{j=1}^N$, we infer the posterior distribution over the latent parameters via Bayes' rule:

$$\mathcal{P}(\{c^{(f_i)}\}, \rho^{(f_i)}, \boldsymbol{\eta}^{(f_i)} \mid \mathcal{D}^{(f_i)}) \propto \mathcal{L}(\mathcal{D}^{(f_i)} \mid \{c^{(f_i)}\}, \rho^{(f_i)}, \boldsymbol{\eta}^{(f_i)}) \cdot \mathcal{P}_c(\{c^{(f_i)}\}) \cdot \mathcal{P}_s(\rho^{(f_i)}) \cdot \mathcal{P}_n(\boldsymbol{\eta}^{(f_i)}),$$

where $\mathcal{L}$ denotes the likelihood function induced by the assumed output distribution $\mathcal{P}$.

For a new input $\boldsymbol{\theta}^*$, the model yields a posterior predictive distribution:

$$\mathcal{P}(\hat{\epsilon}^* \mid \boldsymbol{\theta}^*, \mathcal{D}) = \int \mathcal{P}(\hat{\epsilon}^* \mid \boldsymbol{\theta}^*, \boldsymbol{\vartheta})\, \mathcal{P}(\boldsymbol{\vartheta} \mid \mathcal{D})\, d\boldsymbol{\vartheta},$$

where $\boldsymbol{\vartheta}$ denotes the full set of model parameters. The predictive distribution captures both epistemic uncertainty due to limited data and aleatoric uncertainty inherent in the model formulation.

The MF-BPCE surrogate model algorithm maps an input $\boldsymbol{\theta}$ to both $\epsilon_{CNP}^{(f_0)}$ as low-fidelity output and $\epsilon_{Raw}^{(f_n)}$ as high-fidelity outputs via it's posterior predictive:

$$f : \boldsymbol{\theta} \to \mathcal{P}(\hat{\epsilon} \mid \mathcal{D}) \tag{6}$$

3.3 Likelihood Construction

It is worth noting that the emulated formation efficiency $\hat{\epsilon}$ is still different from the expected gravitational wave rate $\hat{y}$. While a more accurate conversion exists in COMPAS, implementing it is not straightforward, so we decided to leave it for future work. In this work, we adopted a simplified conversion using a constant multiplication factor: $\hat{y} = 314.266854 \cdot \hat{\epsilon}$, which is consistent with community standards. Details regarding this conversion factor can be found in Appendix D.

We then constructed a likelihood function with both $\hat{y}$ and y_{obs}. Because the observed number of events in a fixed volume and time interval is a discrete count, and the events are assumed to occur independently and rarely, the Poisson distribution is the natural and widely adopted model in the astroparticle physics community for such data [Cousins, 2018]. Thus, we define the log-likelihood as:

$$\ln \mathcal{L}(\boldsymbol{\theta}) = \sum_i \left(y_{obs,i} \ln \hat{y}_i(\boldsymbol{\theta}) - \hat{y}_i(\boldsymbol{\theta}) - \ln y_{obs,i}!\right) \tag{7}$$

To infer the posterior over the model parameters $\boldsymbol{\theta}$, we integrate the Poisson likelihood over the surrogate model output $\hat{y}$, and over the latent model parameters $\boldsymbol{\vartheta}$:

$$\mathcal{P}(\boldsymbol{\theta} \mid y_{obs}, \mathcal{D}) = \int\int \mathcal{L}(y_{obs} \mid \hat{y}) \mathcal{P}(\hat{y} \mid \boldsymbol{\theta}, \boldsymbol{\vartheta})\, \mathcal{P}(\boldsymbol{\vartheta} \mid \mathcal{D})\, d\hat{y}\, d\boldsymbol{\vartheta},$$

This formulation reflects both epistemic uncertainty (via $\boldsymbol{\vartheta}$) and aleatoric uncertainty (via $\hat{y}$) in the final inference over $\boldsymbol{\theta}$.

4 Experiment and Validation

Given that the likelihood function relies primarily on $\hat{\epsilon}$, it is crucial to validate that the surrogate model produces accurate estimates of $\hat{\epsilon} \pm \hat{\sigma}$. This section presents experimental results benchmarking RESOLVE against other surrogate models on two different datasets. The three models include **MFGP**—the Multi-Fidelity Gaussian Process surrogate model without the use of Conditional Neural Process (CNP); **RESuM**—the Rare Event Surrogate Model proposed in Schuetz et al. [2025], including both MFGP and the CNP; and the **RESOLVE** model proposed in this work, which combines the CNP with the MF-BPCE algorithm described in Section 3.2. The two datasets include the LEGEND detector design dataset and the BBH merger dataset. Details on the concrete implementation and diagnostics are provided in Appendix H.

4.1 Benchmarking Surrogate Models: Coverage and Accuracy Across Datasets

The LEGEND detector design dataset was provided by Schuetz et al. [2025] with a similar HF/LF structure. The overall goal is to surrogate the design metric $\hat{y} = m/N$ from given design parameter $\boldsymbol{\theta}$. More details about this dataset are provided in Appendix C and the original paper. Benchmarking was performed on a separate validation dataset with 100 out-of-sample HF simulation trials. Each model was benchmarked with four metrics: the mean-squared error (MSE), calculated by averaging $(\hat{y} - y_{Raw})^2$ over the 100 trials, and the 1σ, 2σ, and 3σ coverage, calculated by counting the percentage of trials where y_{Raw} falls within $\hat{y} \pm 1\hat{\sigma}$, $\hat{y} \pm 2\hat{\sigma}$, and $\hat{y} \pm 3\hat{\sigma}$, respectively. As shown in Table 1, the MFGP algorithm (Trial 1) is significantly undercovered. The RESuM model (Trial 2) and the RESOLVE model (Trial 3) both achieve proper statistical coverage on the LEGEND dataset, while RESOLVE outperforms RESuM on the MSE metric.

The major scope of this work is the BBH merger dataset we generated from scratch with COMPAS v03.10.05. For each simulated binary system in COMPAS, users can specify 27 input parameters listed in Appendix B. In this work, we selected four to form the physics parameters of interest $\boldsymbol{\theta}$: the $metallicity$ parameter Z, which sets the initial metal fraction of the progenitor stars; the common-envelope efficiency $\alpha_{\rm CE}$, which governs the orbital shrinkage during unstable mass transfer; the $\sigma_{\rm BH}$ parameter, representing the dispersion of the Maxwellian natal-kick distribution for black holes; and the $\sigma_{\rm NS}$ parameter, defined analogously for neutron stars. The other 23 parameters are treated as nuisance parameters $\boldsymbol{\phi}$.

We then generated HF and LF simulation trials under different $\boldsymbol{\theta}$. Each HF trial cost ~ 4 CPU hours to simulate 10^6 binary systems, while each LF trial cost ~ 15 seconds to simulate 10^3 systems. Within each trial, the four $\boldsymbol{\theta}$ parameters were fixed, while the 23 nuisance parameters varied randomly among binaries. The raw formation efficiency was defined as $\epsilon_{\rm Raw} = m/N$, as described earlier. The training dataset contained 1000 LF trials and 15 HF trials, while the validation dataset contained 200 out-of-sample HF trials.

Table 1: Benchmarking Result on both the LEGEND Detector Design Dataset and the BBH Dataset. For the RESOLVE model, the (d=1/2/3/4/5) in parenthesis means the choice of $1^{st}/2^{nd}/3^{rd}/4^{th}/5^{th}$ order polynomial of the Polynomial Chaos Expansion algorithm, respectively.

Trial	Model	Dataset	(#LF,#HF)	MSE	$1\hat{\sigma}$ [%]	$2\hat{\sigma}$ [%]	$3\hat{\sigma}$ [%]
1	MFGP Schuetz et al. [2025]	Detector Design	(310,10)	0.015	17%	33%	48%
2	RESuM Schuetz et al. [2025]	Detector Design	(310,10)	0.002	69%	95%	100%
3	RESOLVE	Detector Design	(310,10)	0.001	74%	94%	99%
4	MFGP	Black Hole	(1000,5)	9.3×10^{-6}	72.0%	90.5%	96.5%
5	MFGP	Black Hole	(1000,10)	8.1×10^{-6}	68.5%	91.0%	96.0%
6	MFGP	Black Hole	(1000,15)	9.3×10^{-6}	64.5%	88.5%	95.0%
7	RESuM	Black Hole	(1000,5)	17.7×10^{-6}	49.5%	71.0%	78.5%
8	RESuM	Black Hole	(1000,10)	9.2×10^{-6}	51.0%	78.0%	85.5%
9	RESuM	Black Hole	(1000,15)	3.7×10^{-6}	69.0%	92.0%	98.0%
10	RESOLVE (d=1)	Black Hole	(1000,15)	15.3×10^{-6}	66.0%	94.5%	99.0%
11	RESOLVE (d=2)	Black Hole	(1000,15)	7.0×10^{-6}	72.0%	96.0%	98.5%
12	RESOLVE (d=4)	Black Hole	(1000,15)	5.0×10^{-6}	77.5%	96.0%	98.5%
13	RESOLVE (d=5)	Black Hole	(1000,15)	5.3×10^{-6}	76.0%	94.5%	98.5%
14	RESOLVE (d=3)	Black Hole	(1000,5)	18.6×10^{-6}	54.0%	75.0%	86.5%
15	RESOLVE (d=3)	Black Hole	(1000,10)	8.2×10^{-6}	70.5%	91.0%	97.0%
16	RESOLVE (d=3)	Black Hole	(1000,15)	5.3×10^{-6}	76.5%	93.5%	98.5%
	Proper Statistical Coverage				68.27%	95.45%	99.73%

The benchmarking metrics for the BBH datasets are analogous to those of the LEGEND detector design dataset. The results are shown in Table 1. Each model was trained with 1000 LF trials along with 5/10/15 HF trials. As shown in Trials 4-6, the MFGP model without CNP exhibits significant overcoverage in its predictions. The RESuM Model (MFGP+CNP) in Trials 7-9 reached proper coverage at 1σ but is undercovered at 2σ and 3σ. RESOLVE (Trials 10-16) was the only model that achieved proper statistical coverage on BBH dataset. We further examined different choices of polynomial order in the Polynomial Chaos Expansion algorithm of RESOLVE, finding that 3rd order

polynomials yield optimal results. The prediction accuracy and coverage results for each individual trials are illustrated in Figure 2.

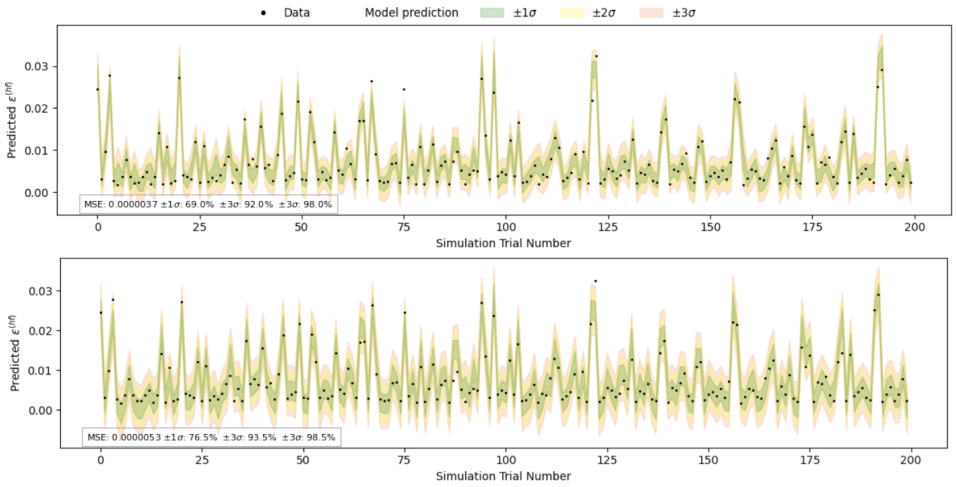


Figure 2: Coverage plot of the RESuM (top) and RESOLVE (bottom) model predictions on the BBH dataset (Rows 9 and 16 in Table 1).

Table 1 summarizes the performance of the three regression frameworks across different HF sample budgets. Several key findings emerge. The MFGP baseline exhibits the highest mean-squared error (MSE) and, strikingly, shows little to no improvement as additional HF samples are added. This suggests that the GP struggles in this setting to extract useful signal from the noisy LF data and may be overly sensitive to the limited HF calibration. By contrast, both RESuM and RESOLVE achieve substantially lower MSE and recover astrophysically sensible parameter dependencies. Their differences become apparent in the coverage properties of the predictive intervals. RESuM attains almost perfect calibration at the 1σ level, but exhibits a clear undercoverage at 2σ (failing to fully capture the tails of the predictive distribution) and a slight undercoverage at 3σ. In contrast, RESOLVE (using $d = 3$ PCE, as lower-order expansions are inconsistent) shows the opposite trend: a marked overcoverage at 1σ (indicating that the emulators are conservative at small scales), essentially perfect coverage at 2σ, and only a mild undercoverage at 3σ. Importantly, however, the gains in reduced MSE only appear once the Conditional Neural Process (CNP) is introduced as a noise mitigation layer: without the CNP, both regression approaches fail to recover consistent parameter dependencies. This demonstrates that denoising the LF input space is a necessary prerequisite for successful multi-fidelity emulation of merger-rate predictions. In practice, the CNP thus plays a central role in enabling both the GP-based RESuM and the PCE-based RESOLVE models to achieve competitive performance in a regime dominated by sparse HF data and noisy LF simulations.

The Bayesian polynomial chaos expansions (as in RESOLVE) offer a particularly attractive framework for our setting. Their global basis representation efficiently exploits even very small numbers of HF simulations, yielding smooth, data-efficient surrogates. More importantly, PCE allows for direct incorporation of physics-motivated constraints—such as boundedness of efficiencies ($0 \leq \epsilon \leq 1$), monotonicity, or asymmetry—through priors on expansion coefficients. This not only improves extrapolation behavior near the edges of the parameter space, but also leads to uncertainty bands that more faithfully reflect the skewed, non-Gaussian distributions characteristic of Poisson-like merger rates. By contrast, Gaussian-process surrogates (e.g., RESuM) excel in modeling local nonlinear features but rely on Gaussian residual assumptions and require larger HF sample sizes to achieve calibrated coverage, which is often impractical in rare-event astrophysical applications. While both approaches perform competitively in terms of mean-squared error, the PCE framework provides clearer analytical interpretability, faster Bayesian inference, and a natural pathway to embed domain knowledge. In this sense, RESOLVE is not only competitive in predictive accuracy but better aligned with the goals of astrophysical population synthesis, where physical consistency and interpretability are as important as raw fit quality.

4.2 Parameter Dependence of the BBH Merger Rate: Comparison with Population Synthesis Studies

A central goal of our analysis is to test whether regression-based emulators can recover the astrophysically expected dependencies of the BBH merger rate on key binary-evolution parameters. Figure 3 shows the marginalized distributions learned by three regression schemes (MFGP, RESuM, and MF-PCE). Comparing the raw LF simulations with their CNP-denoised counterparts highlights that it is essential to suppress stochastic noise and uncover the underlying parameter trends. All three regression approaches then capture the dominant qualitative dependencies, but the global basis of MF-PCE in particular allows it to robustly recover smooth, low-amplitude features even in sparse regions of the training data. Importantly, the subtle correlations learned by RESOLVE are not only statistically significant but also astrophysically interpretable, lending confidence that the emulators are extracting meaningful signal rather than noise.

The combined dependence on progenitor metallicity, common-envelope (CE) efficiency, and natal kicks illustrates both the strengths of the regression models and the astrophysical structure of the parameter space. The merger rate is found to be strongly metallicity-dependent, with the highest efficiencies at very low metallicity ($Z \lesssim 0.005$) and a monotonic decline toward solar values. This trend is robust to variations in CE efficiency and natal kicks, confirming metallicity as the dominant driver of BBH formation efficiency. Such behavior agrees with previous population synthesis studies (e.g., Belczynski et al. 2010, Mapelli et al. 2017, Broekgaarden et al. 2021), which likewise attribute enhanced BBH production in metal-poor environments to weaker stellar winds and correspondingly higher black hole masses. Unlike those works, however, our calculation neglects the metallicity distribution of cosmic star formation and therefore isolates the *intrinsic* efficiency of BBH production. This separation provides a complementary perspective that disentangles stellar-evolution physics from cosmological uncertainties.

The influence of CE physics appears as a second-order modulation on top of the metallicity trend. Population synthesis studies suggest a non-monotonic dependence on the efficiency parameter $\alpha_{\rm CE}$, with merger rates maximized at intermediate values ($\alpha_{\rm CE} \sim 1$–3). Our emulators recover this expected behavior: very low $\alpha_{\rm CE}$ leads to premature coalescence during the CE phase, while very high values produce post-CE binaries that are too wide to merge within a Hubble time [Dominik et al., 2012, 2013, Belczynski et al., 2016, Mapelli et al., 2017, Broekgaarden et al., 2021]. The sensitivity of this channel is itself metallicity-dependent. At low metallicity, efficient BBH production renders the merger rate relatively insensitive to the exact CE efficiency, whereas at higher metallicity, successful BBH formation hinges on a finely tuned $\alpha_{\rm CE}$ value. In this regime, CE physics effectively acts as a gatekeeper, amplifying the already strong metallicity suppression.

Natal kicks provide a third axis of uncertainty. In isolation, increasing the dispersion $\sigma_{\rm kick}$ reduces the merger rate by unbinding binaries at black-hole formation, with the critical scale set by the pre-supernova orbital velocity. A mild compensating effect can occur at intermediate kicks, where the surviving systems are hardened and inspiral times shortened, but for BBHs this contribution is subdominant [Dominik et al., 2012, Fryer et al., 2012, Belczynski et al., 2016, Broekgaarden et al., 2021]. Our models reproduce this monotonic suppression and further reveal how the impact of natal kicks interacts with metallicity and CE physics. At low metallicity, where fallback is strong and direct collapse is common, BBH formation is largely insensitive to kick assumptions. At higher metallicity, where reduced core masses and weaker fallback prevail, survival becomes more sensitive to kick dispersion. Likewise, in evolutionary channels where CE ejection already requires finely tuned conditions, even moderate kicks can suppress the rate. Thus, metallicity defines the baseline BBH production efficiency, CE physics modulates the survival probability of progenitors, and natal kicks act as a final filter on the population.

5 Bayesian Inference and Credible Interval

In Section 4, we demonstrated that RESOLVE produces accurate estimations of formation efficiency with robust statistical coverage while capturing subtle trends across all parameters. Given that the LIGO-Virgo collaboration reported an observed gravitational wave rate of $y_{obs} = 17^{+10}_{-6.7}$ yr^{-1} Gpc^{-3} in Abbott et al. [2023], we ran Bayesian inference to obtain credible intervals for the four physics parameters in $\boldsymbol{\theta}$. The MF-BPCE algorithm in RESOLVE generates a Bayesian posterior during training, which can be directly used as the prior for the Bayesian inference. This approach inherently

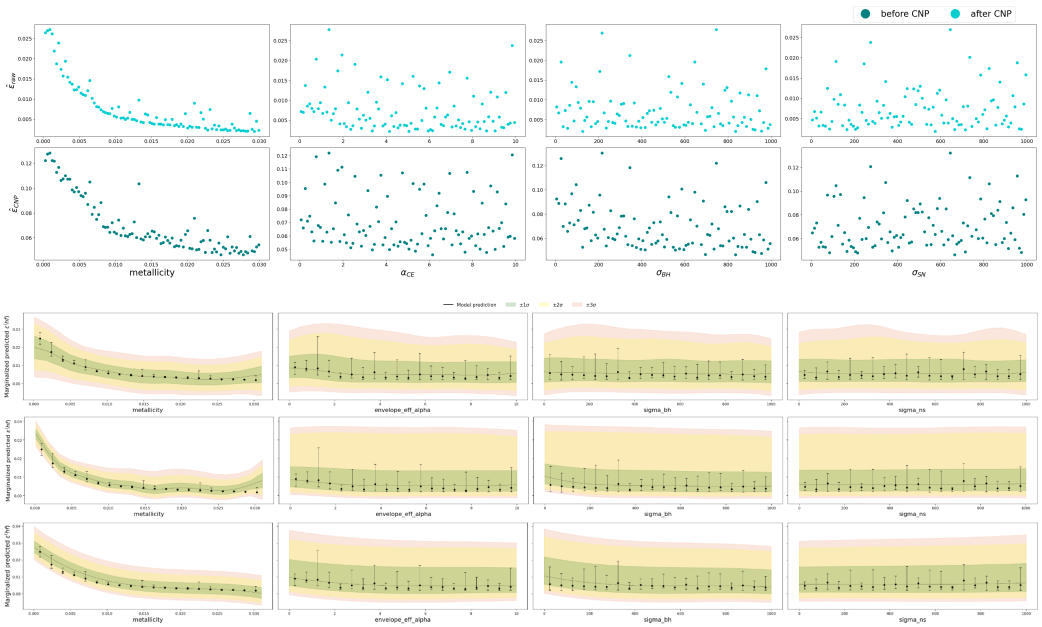


Figure 3: Marginalized parameter dependencies across surrogate models. Row 1 & 2: LF raw simulation outputs and CNP estimetes; Row 3: HF MFGP prediction; Row 4: HF RESuM predictions; Row 5: HF RESOLVE predictions. Panels show dependencies on metallicity (Z), common-envelope efficiency ($\alpha_{\rm CE}$), black-hole natal kick dispersion ($\sigma_{\rm BH}$), and neutron-star natal kick dispersion ($\sigma_{\rm NS}$). Shaded regions denote 1σ, 2σ, and 3σ predictive intervals.

incorporating uncertainty within each parameter of $\boldsymbol{\theta}$. The Bayesian inference uses MCMC algorithm to sample the $\boldsymbol{\theta}$ space where the likelihood functions is minimized. Additional details on our Bayesian inference methodology can be found in Appendix F.

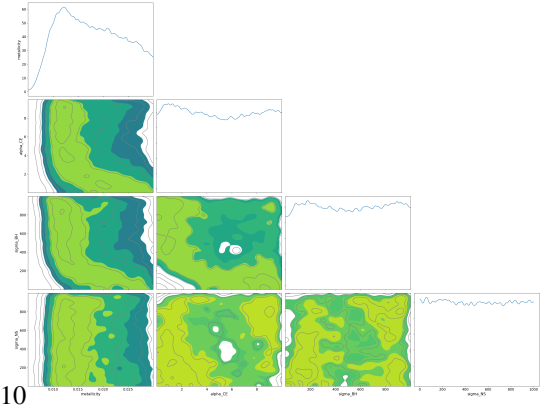
The results of our Bayesian inference are presented in Figure 4. Based on the 17 gravitational wave observations, the inference placed a strong constraint on the metallicity parameter with small uncertainties. This precision stems from the strong and clear dependency between metallicity and formation efficiency, leading to a median metallicity value of $0.0172^{+0.0079}_{-0.0062}$. In contrast, for the remaining three parameters, our model cannot place any strong constraint due to their subtle relationships with formation efficiency, thereby the uncertainty is large.

6 Limitations and Applications

Limitations: This work has three limitations: (1) the relationship $\hat{y} = 314.266854 \cdot \hat{\epsilon}$ is over-simplified. COMPAS has a more sophisticated Cosmic Integration processer to provide a better mapping from $\hat{\epsilon}$ to $\hat{y}$. (2) The Bayesian inference described in Section 5 failed to draw tight conclusions on three parameters beyond metallicity.(3) The current high-fidelity (HF) simulation which requires $\mathcal{O}(4h)$ computation time is still not the most accurate simulation for BBH. A more complex exist but would require $\mathcal{O}(month)$ to generate.

Application: RESOLVE's capability to learn subtle but important trends and provide credible intervals for user-selected parameters of interest could help to map out, for the first

Figure 4: Posterior distributions and correlations of metallicity, $\alpha_{\rm CE}$, $\sigma_{\rm BH}$, and $\sigma_{\rm NS}$, inferred with the MF-PCE algorithm and MCMC. Contours show 68% and 95% credible regions.



time, the high-dimensional parameter space of these gravitational-wave paleontology simulations. This will disrupt the field by: (i) effectively testing different variations of cosmological model assumptions (ii) studies related phenomena, including supernovae, enrichment, galaxy evolution, instrumentation design, and cosmology; (iii) discover and quantify important trends in gravitational-wave paleontology simulations such as how the BBH merger formation efficiency depends on metallicity, supernovae physics, mass transfer events, and stellar winds. In the future, we will leverage RESOLVE to explore these research directions to generate more significant scientific results in gravitational-wave paleontology.

7 Conclusion

In this work, we presented RESOLVE, a rare event surrogate likelihood designed for parameter estimation in Gravitational Wave Paleontology. At its core, we created a novel rare event surrogate model based on the Multi-Fidelity Bayesian Polynomial Chaos Expansion (MF-BPCE) algorithm, which efficiently emulates the BBH merger formation efficiency in gravitational-wave paleontology simulations. We benchmarked RESOLVE against other surrogate models on simulated out-of-sample data, the result shows that RESOLVE is the only surrogate model that successfully approximates gravitational-wave paleontology of BBH with proper statistical coverage. By evolving the BBH merger formation efficiency into a likelihood function, we produced a credible intervals for key astrophysical parameters including metallicity, envelope efficiency, σ_{BH}, and σ_{NS}. These credible intervals can be used by astronomers to validate existing theories or develop new ones regarding the formation, lives, and deaths of stars across cosmic time and the pivotal role they play in shaping our Universe.

Given that high dimensionality and prohibitive simulation costs represent the two principal challenges in Gravitational Wave Paleontology, RESOLVE establishes a framework to address both problems simultaneously. Our future work will proceed along two complementary directions: first, we aim to enhance RESOLVE by systematically addressing the limitations discussed in Section 6; second, we plan to apply the refined RESOLVE framework to understand other properties of binary systems. This work demonstrated how AI can yield scientific conclusions that can be directly used by the astronomical community, fostering a productive interdisciplinary synergy that advances both fields.

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A COMPAS Binary Population Synthesis

We use the open-source Compact Object Mergers: Population Astrophysics and Statistics (COMPAS)[2] binary population synthesis code to simulate the lives of massive binary stars from birth to death. The primary scientific objective of COMPAS is to understand the black holes (BHs) and neutron stars (NSs) whose mergers emit the gravitational waves that are detected today. By simulating the lives of large populations of massive stars, the software is used to forward model observed compact object mergers, helping to constrain the population of compact objects, their massive stellar progenitors, and the physics that underlies stellar and binary evolution.

COMPAS models binary events using simple parametrized prescriptions for isolated stellar and binary evolution, allowing it to compute the complete evolution of a binary in ~ 10 milliseconds on a typical laptop. Other popular codes that use similar approaches to population synthesis include (but are not limited to) MOBSE [Giacobbo and Mapelli, 2018, Giacobbo et al., 2018], COSMIC [Breivik et al., 2020], StarTrack [Belczynski et al., 2002, 2008, 2020], and POSYDON [Fragos et al., 2023, Andrews et al., 2024]. Riley, J. et al. [2022] gives an in-depth description of COMPAS's methodology and implementation, which we describe below.

Single stars constantly undergo nuclear fusion as they evolve, driving changes to their properties and structure, causing phenomena like stellar winds, and triggering events like supernovae. To handle stellar evolution, COMPAS uses the evolutionary formulae for properties including mass, metallicity, radius, and luminosity from Hurley et al. [2000] which are based on stellar models from Pols et al. [1998]. The stellar tracks provided by Hurley et al. [2000] segment the evolution of massive stars according to their evolutionary phases like the main sequence, Hertzsprung gap, giant branch, and more. Once stars run out of thermonuclear fuel, COMPAS uses a variety of class-specific prescriptions to model how they undergo a supernova and leave behind a stellar remnant either consisting of a white dwarf, NS, or BH based on their pre-supernova mass. Stellar remnant evolution is then handled using separate sets of formulae for each remnant type.

As binaries evolve, they often interact through means beyond gravitational attraction, leading to changes in the component properties and orbit. COMPAS parametrizes binary events using their component masses, separation, and eccentricity, and uses prescriptions to account for mass loss, stable mass transfer, unstable mass transfer, supernovae, and stellar contact or mergers. The impacts of gravitational radiation are only considered by COMPAS after both binary components are compact objects, at which point binary eccentricity and separation decrease according to the point-mass approximation from Peters [1964]. If gravitational radiation emission causes a binary to inspiral before Hubble Time (the current age of the Universe), it is considered a merger event.

To evolve populations of binaries, users provide COMPAS with distributions for initial binary properties including component masses, separation, eccentricity, and metallicity. The distributions of initial binary properties are meant to reflect what is observed in the Universe so that users can synthesize populations that represent nature and give insight into sampled subsets. COMPAS

[2] compas.science. Code available at https://github.com/TeamCOMPAS/COMPAS.

also provides several postprocessing scripts for calculating important astrophysical quantities using simulation results. Some of the metrics that COMPAS scripts can calculate include the efficiency of compact object formation and the rates of compact object coalescence and gravitational wave detection.

B COMPAS Training Set Parameter Choices

In this section, the specific parameters for $\boldsymbol{\theta}$ and $\boldsymbol{\phi}$ chosen from the parameter space of COMPAS is outlined. The tables below gives an overview of the physical parameters of interest for $\boldsymbol{\theta}$. The list of parameters chosen for $\boldsymbol{\phi}$ are:

CH on MS(1)	CH on MS(2)	Eccentricity@ZAMS
Equilibrated At Birth	Evolution Status	Mass@ZAMS(1)
Mass@ZAMS(2)	Merger	Merger at Birth
Metallicity@ZAMS(1)	Metallicity@ZAMS(2)	Omega@ZAMS(1)
Omega@ZAMS(2)	PO CE Alpha	PO LBV Factor
PO Sigma Kick CCSN BH	PO Sigma Kick CCSN NS	PO Sigma Kick
ECSN	PO Sigma Kick USSN	PO WR Factor
SN Kick Magnitude Random N	SemiMajorAxis@ZAMS	Stellar Type(1)
Stellar Type(2)	Stellar Type@ZAMS(1)	Stellar Type@ZAMS(2)

C The LEGEND Detector Design Dataset

The LEGEND detector design dataset is curated to find the optimal design of a neutron moderator that slows down and blocks external neutron background from entering the sensitive region. Given a specific detector design, the dataset runs GEANT4 simulations with N neutrons with random initial energy, position and momentum, and count the number of neutrons that enter the sensitive region as m. The design metric is calculated as $y_{Raw} = m/N$. The parameter of interest $\boldsymbol{\theta}$ include 5 design parameters that control the shape of the neutron moderator, while the nusiance parameters $\boldsymbol{\phi}$ are the initial energy, position and momentum of each simulated neutrons.

This dataset contains both HF and LF simulations. The primary difference between HF and LF are (1) the number of neutrons simulated and (2) the physics mechanism behind neutron production. Each HF simulation trial requires 170 CPU hours, while each LF simulation trial only requires 15 CPU minutes. The training datasets contains 310 LF simulation trials and 10 HF simulation trials, while the validation dataset contains 100 out-of-sample HF simulation trials.

After training, the surrogate model will predict $\hat{y} \pm \hat{\sigma}$ from the given $\boldsymbol{\theta}$ of each out-of-sample HF simulations. The means square error is calculated by averaging $(\hat{y} - y_{Raw})^2$ over the 100 trials, while the 1/2/3σ coverage is calculated by counting the number of trials where y_{Raw} fall within $\hat{y} \pm 1/2/3 \times \hat{\sigma}$, respectively. As shown in Table 1, the MFGP algorithm (Trial 1) is significantly undercovered. The RESuM model (Trial 2) and the RESOLVE model (Trial 3) both achieved proper statistical coverage on the LEGEND detector design dataset, while RESOLVE outperforms RESuM on the MSE metric. This means that that RESOLVE achieves better prediction accuracy while maintaining proper statistical coverage.

D Converting Formation Efficiency to Gravitational Wave Rate

Note that the emulated formation efficiency $\hat{\epsilon}$ represents the number of BBH collisions given N simulated binary star systems. This unitless efficiency is different from the expected event rate $\hat{y}$, which has the the unit of number of expected collisions per unit of co-moving volume per year. To obtain $\hat{y}$, we need to perform another step of conversion listed in the following equation:

$$\hat{y} = s\epsilon_r\hat{\epsilon} \tag{8}$$

Where s is the local star formation rate density, which is 1×10^7 times the mass of the sun per co-moving cubic Gigaparsec per year [Madau and Dickinson, 2014]; ϵ_r is a reference efficiency of the mean mass evolved in our simulations which equals $1.57133427 \times 10^{-6}/0.005$ events per solar mass based on [Riley, J. et al., 2022]. Since both numbers are constant, Equation 8 simplifies to a constant number multiplication: $\hat{y} = 314.266854 \cdot \hat{\epsilon}$.

E A Brief Summary of Gravitational Radiation from Rotating events

Following the derivation from Weinberg Steven Weinberg [1972], we consider a stellar body with a mass density denoted by $\rho(\boldsymbol{x'})$. Assuming this body is rigidly rotating about a fixed axis with a constant angular velocity, the power radiated from this event is given by equation (10.5.22) Steven Weinberg [1972]:

$$P(2\Omega) \propto \Omega^6 I^2 e^2 \tag{9}$$

Where Ω is the rotation frequency. Let I_{ij} $(i, j \in \{1, 2, 3\})$ be the moment of inertia tensor for this event. Then I shown above is given by $I = I_{11} + I_{22}$ and:

$$e = \frac{I_{11} - I_{22}}{I} \tag{10}$$

Now, if the rigid body is simply rotating at the center of the rotation axis, then $e = 0$, and thus the radiated power is 0. However, for a event where a body is rotating about the center axis (thus the circular symmetry is broken), both I and e are non-zero, and there is therefore non-zero radiated power.

A event such as a BBH event may admit gravitational waves whereas a single-star event may not, even if it is rotating. Non-zero gravitational wave contributions may arise from particle collisions inside the stellar body, but these effects may pale in comparison to the radiation caused by a binary event.

F Bayesian Markov Chain Monte Carlo (MCMC)

In particle physics and other experimental contexts, Bayesian inference is oftentimes employed to compare data to theoretical models. In this section, we give a brief review of Bayesian Markov Chain Monte Carlo (MCMC) and outline an example of its application to particle physics.

Bayesian statistics, as compared with *frequentist* statistics, uses data from an experiment and a model to give a likelihood as to whether the data matches the model. The method revolves around *Bayes' Rule*. When there are n discrete possible outcomes, this is John Pitman [1993]:

$$P(B_i|A) = \frac{P(A|B_i)P(B_i)}{\sum_i^n P(A|B_i)P(B_i)} \tag{11}$$

This principle, which describes how conditional probabilities may be "flipped" can be easily extended to a statement on theory and experiment. Let M be a theoretical model with a set of model parameters θ_M. This model is to characterize a collection of data, denoted D. The probability that the model parameters are correct given some data and a model is given by Speagle [2020]:

$$P(\boldsymbol{\theta}) \coloneqq P(\boldsymbol{\theta}_M|\boldsymbol{D}, M) = \frac{P(\boldsymbol{D}|\theta_M, M)P(\boldsymbol{\theta}_M|M)}{\int_{\theta_M} P(\boldsymbol{D}|\theta_M)P(\boldsymbol{\theta}_M|M)d\theta_M} \tag{12}$$

For completion, $P(\boldsymbol{D}|\theta_M, M)$ is the probability that the data came from a model with a chosen set of parameters, and $P(\boldsymbol{\theta}_M|M)$ is the probability that the model parameters are true given a model.

In the Bayesian language, one may want to maximize the probability that the model parameters (and the model) accurately describe the experimental data. However, since the "true" model parameters may not be known, a surrogate scenario may be used, where the *expected loss* may be minimized. That is, if the true value of $\boldsymbol{\theta}$ is known, different values of parameters may be suggested, $\hat{\boldsymbol{\theta}}$, and a *loss function*, $L_P(\hat{\theta}, \theta)$, can penalize this suggestion by comparing with the true value. However, since the true parameter might not be known, one can marginalize over all possible parameters by:

$$L_P(\hat{\theta}) \coloneqq \mathbb{E}_P[L_P(\hat{\theta}|\theta)] \tag{13}$$

And find the best value of $\boldsymbol{\theta}$ by minimizing the expected loss:

$$\underset{\hat{\theta}\in\theta}{\arg\min}\ \mathbb{E}_P[L_P(\hat{\theta}|\theta)] \tag{14}$$

While this may be formally straightforward, it is oftentimes computationally expensive to navigate over all possible model parameters, and this problem can scale as an NP[3]-complex problem.

However, there are ways to improve the complexity of the problem. One way is to traverse the joint parameter space at (potentially) uneven intervals. That is, the expected loss may be approximated by a discrete sum that resembles a Riemannian sum:

$$\mathbb{E}_P[L_P(\hat{\theta}|\theta)] = \int L(\boldsymbol{\theta})P(\boldsymbol{\theta})d\theta \rightarrow \sum_{i=1}^{n} L(\boldsymbol{\theta}_i)P(\boldsymbol{\theta}_i)\Delta\boldsymbol{\theta}_{i,i+1} \tag{15}$$

Where the term $\Delta\boldsymbol{\theta}_{i,i+1}$ denotes the spacing between the set of n model parameters. It can be shown Wiegand [1968] that the most optimal way to space the intervals is such that the corresponding spacing on the posterior distribution, $P(\boldsymbol{\theta})$, is minimal where the posterior is greatest in values and maximal where it is lowest. In this way, the model parameters that marginally contribute the least to the posterior distribution are "skipped over" the fastest, and the parameters that contribute the most are tediously searched over. Stated pictorially, this is:

$$\text{parameter spacing} \propto \frac{1}{\text{posterior distribution}} \rightarrow Q(\boldsymbol{\theta}) \propto \frac{1}{\Delta\boldsymbol{\theta}(\boldsymbol{\theta})} \tag{16}$$

This strongly motivates (and exactly produces) a new distribution, called the *proposal distribution*, $Q(\boldsymbol{\theta})$. In the background, this has changed the parameter search problem introduced in Equation 14 to be:

$$\mathbb{E}_P[\,\cdot\,] \rightarrow \mathbb{E}_Q[\,\cdot\,] \tag{17}$$

Using this formalism, the problem statement shifts to a problem regarding the distribution of parameter space rather than the posterior distribution itself. *Monte Carlo methods*, which are computational techniques involving sampling from a discrete set of values as a stand-in for the distribution in question, can now be employed on the distribution of possible intervals.

Having motivated the use of Monte Carlo methods in searching parameter spaces in Bayesian inference problems, we discuss the Markov Chain Monte Carlo technique. This approach seeks to create an optimal distribution over parameter space by generating samples from a Markov Process. A Markov Process, in this context, is a random walk in parameter space that produces a set of parameters explored. That is, the *chain* of parameters explored is described by n random steps (iterations):

$$\{\boldsymbol{\theta}_1, ..., \boldsymbol{\theta}_n\} \tag{18}$$

From this chain of parameter values, a probability density is mimicked by simply counting the number of parameter values from the chain inside a volume element and dividing by the total number of values generated: note that this volume element is the same used in computing the posterior distribution. This equation is given in Equation 44 in Reference Speagle [2020]:

$$\int_{\theta\in\delta_\theta} P(\boldsymbol{\theta})d\theta \approx n^{-1}\sum_{j=1}^{n} \mathbb{1}[\boldsymbol{\theta_j} \in \delta_\theta] \tag{19}$$

In the above equation, $\mathbb{1}[\,\cdot\,]$ is the indicator function and δ_θ is the volume element. Since this stand-in density is inversely proportional to the posterior distribution, simulation can now be made over parameter space instead of the posterior using a Markov Chain process. One popular implementation of a Markov Chain Monte Carlo technique is the Metropolis-Hastings Algorithm Metropolis et al. [1953]. In this method, the chain of parameters is produced (at least conceptually) in the following way:

1. In the chain of parameters, suggest a new parameter based on the current parameter in the list using a *proposal distribution*, $Q(\boldsymbol{\theta_{i+1}}|\boldsymbol{\theta_i})$
2. This new parameter is accepted with some transition probability or rejected with a potentially different probability
3. Once the probability of accepting a new parameter is equal to rejecting the parameter[4], the optimal proposal distribution has been found

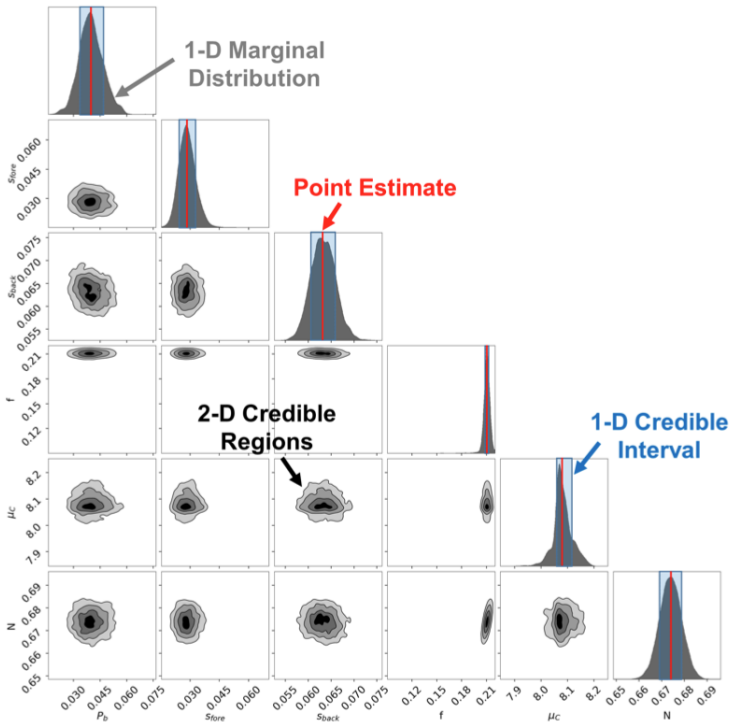
[3]Non-deterministic polynomial time

[4]This idea stems from *detailed balance*, which coincidentally has strong connections in equilibrium statistical mechanics

This proposal distribution emulates the behavior of the density function described by Equation 19, which means it has direct relation to the posterior distribution.

G Bayesian MCMC in Particle Physics

In Appendix F, we described how Bayesian MCMC changes the language of model tuning from one involving the posterior distribution to a a search in parameter space. In this section, we describe how such an approach is commonly used in particle physics applications, and motivate its use in an astronomical setting. Using the principles of Bayesian MCMC, experimental particle physicists test theoretical models by optimizing the model's parameters. However, finding the optimal parameter space distribution is merely the beginning of the analysis. After such distribution is found, what is oftentimes most useful to physicists is the marginal distribution over different parameters, as understanding high-dimensional spaces may not be as intuitive. Consider the following figure from Speagle [2020]:

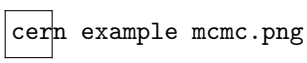


This generic graph shows the marginal distributions over several parameters, namely $\theta = \{P_b, s_{fore}, s_{back}, f, \mu_c, N\}$. Viewing this set of graphs as a matrix, the diagonal elements (shown with red lines) are the 1-dimensional distributions of the parameters: for example, the top left graph is a 1-dimensional marginal distribution of P_s (along the x-axis). The "off-diagonal" graphs are the joint marginal distributions between *two* parameters. This explains why the graphs are shown with contours, as a probability density between two variables is naturally three-dimensional. Filling in all the "off-diagonal" graphs, two-variables at a time may be compared, which has a much easier interpretation for how different model parameters vary together. Having motivated why marginalizing the MCMC parameter distribution gives insight into how model parameters vary together, we now turn to an example of their application in particle physics.

CERN, which is the world's leading high energy particle physics experiment, is home to several important scientific advancements, including the discovery of the Higgs Boson Collaboration [2012]. To explore high energy physics, particles are accelerated to ultra-relativistic[5] speeds and

[5]Speeds close to the speed of light

collide into other particles to expose physics processes at the subatomic scale. This process, broadly described by *scattering theory*, can quickly demand a large amount of model parameters that must be tested experimentally. Many of CERN's experiments must, therefore, explore large parameter spaces to verify the experimental data. MCMC is a natural choice for analyzing experimental data. One example of CERN using MCMC in their analysis may be found from Coccaro et al. [2020]. In this work, a new framework for likelihood estimation in large parameter spaces is presented. Part of the model features sampling from an MCMC process, and in Figure 9 a plot similar to Figure G is shown Coccaro et al. [2020]:



In this figure, marginal distributions are shown for various model parameters in a machine learning model. While this paper is certainly not the only use of MCMC in CERN experiments, it does underscore the basic principles of MCMC in high energy physics. Moreover, CERN is not the only experiment in high energy physics that employs MCMC.

H Model Implementation and Diagnostics

Accurate surrogate modeling in a multi-fidelity setting requires balancing model complexity with the availability of data across fidelity levels. To achieve this, we adopt a targeted polynomial order selection strategy, employ Bayesian inference with carefully chosen priors, and assess model quality through a comprehensive set of diagnostics. All core modeling components are implemented in Python, using the `chaospy`, `scikit-learn`, `PyMC`, and `aesara` libraries.

H.1 Polynomial Order Selection

The polynomial order for the low-fidelity model is selected using K-fold cross-validation. For a predefined set of candidate degrees d, we generate the corresponding multivariate orthonormal polynomial basis using the `chaospy` package and split the low-fidelity training data into K folds. The surrogate model is trained on $K-1$ folds and evaluated on the remaining fold by computing the mean squared error (MSE). This procedure is repeated across all folds, and the average MSE is used to estimate the generalization error for each degree. The optimal degree d^* is then selected as the one minimizing the average cross-validated MSE.

To mitigate overfitting and enforce sparsity in the regression coefficients, we apply LASSO regression with automatic cross-validation using `scikit-learn`'s `LassoCV`. This step helps identify the most relevant polynomial terms for the surrogate expansion.

Due to the sparsity of higher-fidelity data, we fix the polynomial order for the discrepancy model (used to correct the lower-fidelity prediction) to 1. This linear model captures leading-order deviations without overfitting and ensures stable estimation when only a limited number of high-fidelity samples is available.

H.2 Bayesian Inference

Bayesian inference is performed using Hamiltonian Monte Carlo (HMC), specifically the No-U-Turn Sampler (NUTS), implemented via the `PyMC` probabilistic programming framework. This allows for efficient exploration of the high-dimensional posterior space and provides full uncertainty quantification over model parameters. Tensor algebra operations needed in the model specification, such as linear transformations and inner products, are expressed symbolically using `aesara`, which underpins `PyMC`'s computational backend.

Priors are specified individually for each fidelity level, allowing the model to reflect varying levels of uncertainty and data availability across fidelities.

H.3 Prior Distributions

We place prior distributions on the unknown model parameters to encode our assumptions and promote regularization:

- **PCE coefficients:** Each coefficient $c_{j/k}^{(f_i)}$ is assigned a zero-mean Gaussian prior $\mathcal{N}(0, \tau_{j/k}^2)$, encouraging sparsity and penalizing irrelevant terms.
- **Noise variance:** The noise standard deviation $\sigma^{(f_i)}$ is modeled with a Half-Normal prior, $\sigma^{(f_i)} \sim \text{HalfNormal}(\lambda)$, where λ is set to the empirical standard deviation of the high-fidelity data. This reflects the belief that noise is strictly positive and typically small.
- **Fidelity scaling coefficient:** For each fidelity level $f_i >$ LF, we introduce a scaling coefficient $\rho^{(f_i)} \sim \mathcal{N}(\mu, \sigma^2_{\rho^{(f_i)}})$. Rather than fixing the mean $\mu = 1$, we determine it by minimizing the MSE between the lower-fidelity and the higher-fidelity training data. This optimization allows $\rho^{(f_i)}$ to reflect empirical discrepancies between fidelity levels rather than enforcing a fixed prior belief.

These priors are flexible and can be adapted to encode stronger or weaker assumptions, depending on domain knowledge and the observed data characteristics.

H.4 Model Comparison and Diagnostics

To evaluate model quality and support model comparison, we compute the Widely Applicable Information Criterion (WAIC) and Leave-One-Out Cross-Validation (LOO-CV) using Monte Carlo estimates from posterior samples obtained via `PyMC`. These criteria provide insight into out-of-sample predictive performance and are complemented by a robust set of additional diagnostics:

- **Calibration:** Posterior predictive intervals at $\pm 1\sigma$, $\pm 2\sigma$, and $\pm 3\sigma$ are visually inspected to assess coverage and uncertainty calibration.
- **Predictive performance:** We report the mean squared error (MSE), expected log predictive density (elpd_{loo}), its differences across competing models ($\text{elpd}_{\text{diff}}$), and associated standard errors. The effective number of parameters is estimated via p_{loo} and p_{waic}.
- **Empirical coverage:** We compute empirical coverage probabilities at the $\pm 1\sigma$, $\pm 2\sigma$, and $\pm 3\sigma$ levels to evaluate the reliability of posterior intervals.

H.5 Sampling Diagnostics

We monitor the efficiency and convergence of the MCMC sampling process using several diagnostics, available through `PyMC` and its companion library `ArviZ`:

- **Bayesian Fraction of Missing Information (BFMI):** Assesses whether the HMC algorithm adequately explores the energy landscape.
- **Potential scale reduction factor ($\hat{r}$):** Checks for convergence across multiple chains.
- **Effective sample size:** Inferred from posterior variance and stability of interval estimates.

We also examine the Pareto shape parameters and posterior variance of log predictive densities to identify influential observations that may compromise the reliability of WAIC and LOO-CV. These diagnostics help ensure robust inference and guard against overconfidence in model selection based on information criteria alone.

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